

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}\delta_\sigma L_m^i &= -\sigma L_n^i \mathfrak{m}^n_m, \\ \delta_\sigma A^m_\mu &= \sigma \mathfrak{m}^m_n A^n_\mu, \\ \delta_\sigma B_{\mu\nu m} &= -\sigma B_{m\mu\nu} \mathfrak{m}^n_m + 2\eta_{mn} \partial_{[\mu} \sigma A^n_{\nu]},\end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}\mathfrak{Q}_n^{k-1}(x_{[n]}^k, \xi_m) &= F_{\tau_m}(x_{[m]}^k, \Psi_m) - \mathcal{Q}_m(x_{[m]}^k) + \mathcal{Q}_m^{k-1}(x_{[m]}^k) \\ &\geq \mathfrak{Q}_n(x_{[n]}^k, \xi_m) - \mathcal{Q}_m(x_{[m]}^k) + \mathcal{Q}_m^{k-1}(x_{[m]}^k)\end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}\left\| \lambda_s(b) \left(\sum_{t \in G} b_t \delta_t \right) \right\|^2 &= \left\| \sum_{t \in G} b b_t \delta_{st} \right\|^2 = \left\| \sum_{t \in G} (b b_t)^* b b_t \right\| = \left\| \sum_{t \in G} b_t^* (b^* b) b_t \right\| \\ &\leq \left\| \sum_{t \in G} \|b\|^2 b_t^* b_t \right\| = \|b\|^2 \left\| \sum_{t \in G} b_t^* b_t \right\| = \|b\|^2 \left\| \sum_{t \in G} b_t \delta_t \right\|^2\end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}\rho(\alpha(x), \alpha(y)) \circ \beta &= \beta \circ \rho(x, y), \\ \rho(\alpha(x), \alpha(y)) \rho(a, b) - \rho(\alpha(a), \alpha(b)) \rho(x, y) &= (\rho([x, y, a], \alpha(b)) - \rho([x, y, b], \alpha(a))) \circ \beta, \\ \rho([x, y, a], \alpha(b)) \circ \beta - \rho(\alpha(y), \alpha(a)) \rho(x, b) &= \rho(\alpha(a), \alpha(x)) \rho(y, b) + \rho(\alpha(x), \alpha(y)) \rho(a, b),\end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}\mu_{\omega, c} &\leq S_{\omega, c}(\lambda_0 \Phi_\omega) \\ &= \lambda_0^2 L(\Phi_\omega) + \lambda_0^3 N(\Phi_\omega) + \lambda_0^2 \omega Q(\Phi_\omega) + \lambda_0^2 c P(\Phi_\omega) \\ &= \lambda_0^2 S_{\omega, 0}(\Phi_\omega) + \lambda_0^2 \{(\lambda_0 - 1) N(\Phi_\omega) + c P(\Phi_\omega)\} \\ &= \lambda_0^3 \mu_{\omega, 0} < 8 \mu_{\omega, 0}.\end{aligned}$$